IT4208: Open source computing

- The basic principles of open source computing,
- Open source technologies,
- Open source development platforms,
- Managing open source software,
- Business model of open source computing,
- Open source software licenses,
- Open source project structure,
- Distributed team software development,
- Current events in the open source world,
- Open source project Participation and contribution,

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Chapter Contents

- Module 1
  - Chapter 1. Introduction
  - Chapter 2: Principles of Open source computing
  - Chapter 3: Opensource Organization and the future of opensource
- Module 2:
  - Chapter 4: Opensource Development Platforms
  - Chapter 5: Running open source software
- Module 3:
  - Chapter 6: Open source business models

Text book :

- Fadi P. Deek and James A.M.McHugh
  - Open Source Technologies and Policies, Cambridge University Press

References:

1. Karl Fogel,
   - Producing Open Source Software
2. Andrew,
   - Understanding Open Source & Free Software Licensing, O'Reily

Learning Objective

- Develop an opensource project using Gitlab open source platform
  - Application level (blooms taxonomy)

Supporting Learning Materials

- Selected open source projects available on the web
- Experience sharing with local industry open source contributors
- Chapter 4, open source development platforms

Bloom's Taxonomy for Educational Objectives (revised)

Action Verbs and Activities

Higher Order Thinking

	Verbs		Activities	
Creating	• Construct • Design • Organize	• Compose • Prepare • Invent	• Articles • Experiments • Case Studies	• Invention • Games • Projects
Evaluating	• Estimate • Prioritize • Test	• Explain • Rank • Defend	• Critiques • Simulations • Appraisals	• Valuing • Surveys • Court Trials
Analyzing	• Contrast • Deconstruct • Question	• Appraise • Criticize • Outline	• Survey • Case Studies • Discussion	• Graph • Argument • Questionnaire
Applying	• Interpret • Experiment • Discover	• Calculate • Operate • Show	• Forecast • Build/Create • Role Play	• List • Painting • Project
Understanding	• Describe • Classify • Summarize	• Report • Identify • Associate	• Oral Report • Diagram • Analogies	• Summary • Photograph • Model
Remembering	• Recall • Locate • Retrieve	• Collect • Define • Label	• Examples • Illustrations • Analogies	• Multiple Choice • Lecture • Visuals/ Audio

Lower Order Thinking

Assignment

- In Module 1, Students brainstorm on developing open source project ideas
- In Module 2, students choose an open source project ideas
- In Module 3, Student submit a developed open source Project

- Continuous Assessment (30%)
- LAB (40%)
- Final 30%

Chapter 1: Introduction to Open Source Computing

What is open source

History and evolution of open source

Open source vs proprietary software

Advantages and myths of open source

Open source technologies

Open source in education, industry, and government

Open Source History

- Open source is ubiquitous
- Open source is what powers the internet
  - A study shows 60% of the world’s websites run on Apache and Nginx, the most popular web servers.
- One of the reasons open source software has massive usage is:
  - There are a lot of contributors
  - Developers from anywhere in the world can bring their ideas, perspectives, and use cases to software, making is more dynamic. This leads to wider adoption.

Open source history …

- 1950s—the A-2 System (equivalent of today's compilers) --released with its source code
- 1980s – GNU, a free operating systems that was made to counter closed systems by Stallman. He also founded Free Software Foundation (FSF),
  - Later Open source initiative (OSI)
  - Python software foundation
  - Apache Software Foundation
- 1990s –Linux. Linux kernel become the GNU operating system
- Python interpreter source code (1991)
- Apache HTTP server (1995)
- Coning of the term 'open source' (1998)
- Netscap browser's source code (1998)
- Release of open office's source code (2000)

Open Source in the 2000s

(2000s -2010s: exponential growth in open source)

- The Free Software Foundation Europe is created to support free software in Europe (2001)
- Version 1.0 of Mozilla Firefox becomes available to the public (2004)
- Git, a version control system, was released in 2005 by Linus Torvalds, Linux's creator.
- Google releases the Android mobile OS (2008)
- Blockchain is built on lots of open source technologies and developers contribute to its code base (2008)
- Node.js comes out (2009)
- Google Chrome is based on the Chromium project's open source code. Chrome was initially released in 2011
- A lot of open source tools power IoT, including the Linux Kernel. IoT became a conglomeration of technologies in 2013
- Containers and container orchestration (Docker 2013 and Kubernetes 2014)
- The cloud (2016)
- Firefox Quantum (2017). Firefox's biggest update in 2017, saw it rise to become the third most used browser.

So, what is an open source

- As its core, open source is an idea. A concept that revolves around access
  - You can access someone else's (or company's) source code. Customize it for your use case find bugs, and fix them too. You can also share 'version' of the source code in addition to the original
- Opens source can also be seen as a movement
  - A 'rebellion' against close source (proprietary software which is not accessible)
  - Use case match issues, bug issues, security, waiting for the next big update
  - Avoiding Vendor lock-in
- But note that, open source is a concept that is applicable to other fields beyond technology

Benefits of Open source vs Proprietary

- Control:
  - You can change the code if needed. You also own the system and can control who has access to it and any data it includes.
- Flexibility:
  - You can customize it to your heart's content. No bound with licensing or usage limits.
- Costs
  - It is free. You only worry for any technical infrastructure such as hosting.
- Ease of Use:
  - They tend to be more turnkey. It is easier than building from a scratch.
- Dedicated Support:
  - Provides a dedicated support for customers.
- All-in-One Subscription:
  - They include hosting,

Open Source Myths

- Despite the growth of OSS,
  - Inferior in quality, security, longevity (long duration of life)

In comparison to proprietary software when there is plenty of evidence to the contrary

Iconic open source projects

- Linux, mozilla firefox, GNU compiler
- Market share: $32 Billion by 2023

1. Myth 1: Open source is less secure than proprietary software
   - Allow for development teams to rapidly address any security issues, whereas proprietary systems must rely on the software vendor to get around to issuing an update or patch
2. Myth 2: Open source is harder to maintain
   - Are being developed, used, and improved by recognized technology giants. And opensource contributions are traceable and well documented, marking them transparent and not very difficult to maintain
3. Myth 3: There is less support for open source software
   - Google and Facebook rely on Linux to run their services and hence concerned with maintaining the software and third party integration than wait for vendor
4. Myth 4: Open source is not enterprise-grade
   - Amazon, Google, IBM, Facebook, Twitter use open source
   - They are not built chaotically, they have strict governance of what’s added.

Recap: Advantages

- Flexiblity and customizability
- Affordability and Auditability
- Quality
- Freedom (unlike vendor lock-in)
- Support

- Towards an open Government
  - Preaches a government which is highly transparent and offers mechanisms for continual public scrutiny
  - Envisions increased citizen participation and collaboration in all government proceedings through the application of modern and open technologies
- If the future is a walk towards an open government, open source technologies will perhaps play the most significant role

Open source Licenses

Who owns Open Source Software ?

- Is there any particular person ?
- Is there any particular group of people
- Ideally the authors of the software are its owners but there are thousands in some (contributors, maintainers, committers)
- “Ownership “ in terms of rights, intellectual property, or copyright

1. MIT License
   - Popular permissive open source license that gives you a copyright license but not a patent one. Allows commercial use of the software
2. Apache 2.0 License
   - Highly permissive license, allows using , distributing, and modifying open source software without paying royalties or infringing patents
3. GNU GPL
   - Less permissive that MIT and Apache. Any subsequent work on the open source software must be distributed under the same terms.

Open Source Technologies/Development Perspective

Open Source Cloud Platforms Leveraged for Cloud Software Development

- Cloud Foundry
- WSO2
- Alibaba Cloud
- OpenShift
- Tsuru
- OpenStack
- Stackato
- Cloudify
- CloudStack
- AppScale

Open Source Tools and Technologies for Managing and Analyzing Big Data

- Hadoop
- Apache Cassandra
- CouchDB
- Kafka
- PostgreSQL
- Apache Spark
- MongoDB
- HPCC Systems
- Redis
- MariaDB

Open Source Frameworks for Artificial Intelligence/ Machine Learning/ Deep Learning

- TensorFlow
- PyTorch
- Scikit-learn
- Theano
- Microsoft Cognitive Toolkit
- Keras
- OpenCV
- Caffe2
- RNN
- Amazon SageMaker Neo

Top Open Source Programming Languages and Frameworks

- Python
- Java
- C++
- Kotlin
- R programming
- Golang (Go)
- C#
- JavaScript
- Ruby
- Swift

JavaScript backend/frontend

- Node.js (backend)
  - A java script server side framework for building web/mobile apps
- React / React Native (frontend)
  - A java script library for building user interfaces.

Active Learning (Graded 10%)

- Group Students in to 5 to brainstorm on state of the art open-source technologies (consider contextualization to Tigray region)
  - Open source in Education
  - Open source in Health
  - Open source in Agriculture
  - Open source in Government
  - Open source in Industry
- Students write a report and present their works.